Full name: Tsai-Hsuan Hsu **Preferred name:** Tony **Location:** Sydney, Australia
Language: Mandarin (native), English (fluent), Japanese (intermediate)
Email: tony.tsaihsuan.hsu@gmail.com LinkedIn GitHub

Education

PhD candidate Oct 2022 – 30 Jun 2026 (thesis submitted)

School of Life and Environmental Sciences, The University of Sydney (USYD), Sydney, Australia

- Thesis: *Drivers and consequences of fish corallivory on coral reefs*
- Supervisors: Prof. Will F. Figueira and Prof. Andrew S. Hoey
- Honours: Recipient of Taiwan & USYD Joint PhD Scholarship (Total value: ~AU$316k)

M.Sc. Sep 2020 – Jun 2022

Marine Biology and Fisheries Division, National Taiwan University (NTU), Taipei, Taiwan

- Thesis: *Navigating the scales of diversity in subtropical and coastal fish assemblages ascertained by eDNA and visual surveys*
- Supervisors: Assoc. Prof. Vianney Denis and Prof. Wei-Jen Chen
- Honours: Dean's Award (Top academic performance)

B.Sc. Sep 2016 – Jan 2020

Department of Life Science, NTU, Taipei, Taiwan

- Honours: Three-time recipient of the prestigious Presidential Award

Research & Professional Experiences

PhD Researcher & Project Contributor Oct 2022 – Present

USYD, Sydney, Australia | The Australian Institute of Marine Science, Townsville, Australia

- Designed and executed comprehensive field surveys investigating reef fish assemblages and feeding behaviours.
- Contributed to *Ecological Intelligence for Reef Restoration Project (EcoRRAP)*, conducting UVC and RUVs.
- Independently managed complex compliance approvals, successfully securing Great Barrier Reef Marine Park Authority (GBRMPA) and Queensland Fisheries permits, alongside animal ethics approvals.
- Supported and mentored junior team members both in the field and in the office.

Casual Academics Jul 2023 – Present

USYD, Sydney, Australia

- Mentored and supervised undergraduate students in laboratory and field settings for *Biostatistics*, *Marine Biology*, and *Coral Reef Biology* courses.
- Evaluated technical reports and provided constructive feedback on data analysis and scientific writing.

Research Assistant Jun 2022 – Aug 2022

Ocean Center, College of Science, NTU, Taipei, Taiwan

- Project: *Important Ecosystems Survey and Ecosystem Services Evaluation in Sea Areas around Taiwan*
- Conducted marine fauna baseline surveys, collected environmental data, and assisted with data management and technical reporting.

Master's researcher Sep 2020 – Jun 2022

Functional Reef Ecology Lab, Institute of Oceanography, NTU, Taipei, Taiwan

- Investigated subtropical reef fish diversity by integrating eDNA with traditional visual & video censuses.

Field & Survey Techniques

Marine fish surveys:
Underwater Visual Census (**UVC**), Unbaited Remote Underwater Videos (**RUVs**), and environmental DNA (**eDNA**).
Benthic Surveys:
Photogrammetry, Photo-quadrat surveys, Coral sampling & tagging.
Scientific Diving:
PADI Rescue Diver (338+).

Lab & Analytical Skills

Molecular: Fish eDNA analysis, DNA extraction, PCR, and Capillary electrophoresis.
Biochemical: Lipid extraction & Fatty acid analyses (GC-MS), Stable isotope analysis (DELTA V Advantage), Protein & chlorophyll extraction.
Morphological: Otolith extraction and Fixation of the fish specimen.

Data Analyses & Software

Statistical Analysis: Advanced statistical analysis and Data visualization using **R software**.
Image Analysis: Video annotations (EventMeasure, SeaGIS) and Photo-quadrat analysis (CoralNet, CPCe).
Bioinformatics: Phylogenetic tree construction (CIPRES, MEGA).
Microsoft Office 365

Research Grants & Scholarships

Taiwan and USYD Joint PhD Scholarship | equiv. AU$ 316k	Oct 2022 – Jun 2026
Student Travel Awards for ICRS, Australian Coral Reef Society | AU$1k	2026
Postgraduate Research Support Scheme (×3), USYD | AU$7.7k	2023–2025
Ruhm Award, Marine Studies Institute, USYD | AU$ 5k	2024
Holsworth Wildlife Research Endowment, Ecological Society of Australia | AU$ 7.4k	2023
One Tree Island Research Station Scholarship, USYD | AU$ 4k	2023
Research Scholarship, Institute of Oceanography, NTU | AU$ 2.4k	2020–2022

Academic Awards & Honours

Honourable Mention Oral Presentation Award, Taiwanese Coral Reef Society	2026
Dean’s Award, College of Science, NTU	2022
First prize, Youth Forum Presentation, Institute of Oceanography, NTU	2022
Excellent Performance Presentation Award, Taiwanese Coral Reef Society	2022
Honourable Mention Poster Award, Taiwanese Coral Reef Society	2020
Presidential Award (×3), Department of Life Science, NTU	2017–2019

Conference presentations

- (Oral) *International Coral Reef Symposium (ICRS),* Auckland, New Zealand — 2026
- (Oral) *Taiwan International Symposium on Coral Reefs*, Kaohsiung, Taiwan — 2026
- (Oral) *HDR Showcase*, SOLES, USYD, Sydney, Australia — 2025
- (Oral) *Joint Conference of ASI and IPFC 12*, Taipei, Taiwan — 2025
- (Oral) *Taiwan International Symposium on Coral Reefs*, Taichung, Taiwan — 2024
- (Poster) *Australian Marine Sciences Association (AMSA)*, Hobart, Australia — 2024
- (Poster) *HDR Showcase*, SOLES, USYD, Sydney, Australia — 2024
- (Oral) *Australian Marine Sciences Association (AMSA)*, Gold Coast, Australia — 2023
- (Oral) *HDR Showcase*, SOLES, USYD, Sydney, Australia — 2023
- (Oral & Poster) *Youth Forum*, IONTU, NTU, Taipei, Taiwan — 2022
- (Oral) *Taiwanese Coral Reef Society*, Taipei, Taiwan — 2022
- (Oral) *The Ichthyological Society of Taiwan*, Taipei, Taiwan — 2021

- (Poster) *Taiwanese Coral Reef Society*, Taipei, Taiwan 2020

Other qualifications

Driving License (NSW, Australia and Taiwan)

General Boat License (NSW, Australia)

First aid, CPR, Advanced resuscitation and O_2 therapy

Publications

7 publications in scientific journals

Hsu, T.-H. T., Hoey, A. S., Ferrari, R., Toor, M., Gordon, S., Remmers, T., Smallhorn-West, J., Piccaluga, A., & Figueira, W. F. (in press). Corallivore assemblage, feeding behaviour, and cover of *Acropora* corals drive fish corallivory on tropical reefs. *Ecosphere*

Hsu, T.-H. T., Hoey, A. S., Warren, C. R., Marrable, I., & Figueira, W. F. (2026). Behavioural, physiological, and biochemical responses of two species of scleractinian coral to butterflyfish predation. *Mar Environ Res*, 218, 108026. https://doi.org/10.1016/j.marenvres.2026.108026

Hsu, T.-H. T., Gordon, S., Ferrari, R., Hoey, A. S., & Figueira, W. F. (2025). Optimizing remote underwater video sampling to quantify relative abundance, richness, and corallivory rates of reef fish. *Coral Reefs*, *44*(2), 435-449. https://doi.org/10.1007/s00338-024-02613-6

De Palmas, S., Chen, Q., Guerbet, A., Hsieh, Y. E., **Hsu, T.-H. T.**, Lin, Y. V., Sturaro, N., Wang, P.-L., & Denis, V. (2025). Energy acquisition and allocation strategies in scleractinian corals: insights from intraspecific trait variability. *Coral Reefs*, *44*(2), 489-500. https://doi.org/10.1007/s00338-025-02617-w

Sommer, B., Malcolm, H. A., Cooney, C., Dalton, S. J., Foo, S. A., Fraser, N., **Hsu, T.-H. T.**, Maguire, D., Nimbs, M. J., Strehling, N., Webb, M., & Byrne, M. (2025). Understanding and managing species range shifts: first observed mixed-species outbreak and coral predation by crown-of-thorns starfish (*Acanthaster brevispinus* and *A.* cf. *solaris*) on subtropical reefs. *Mar Environ Res*, *212*, 107556. https://doi.org/10.1016/j.marenvres.2025.107556

Hsu, T.-H. T., Chen, W.-J., & Denis, V. (2023). Navigating the scales of diversity in subtropical and coastal fish assemblages ascertained by eDNA and visual surveys. *Ecological Indicators*, *148*, 110044. https://doi.org/10.1016/j.ecolind.2023.110044

Hsu, T.-H. T., Carlu, L., Hsieh, Y. E., Lai, T.-Y. A., Wang, C.-W., Huang, C.-Y., Yang, S.-H., Wang, P.-L., Sturaro, N., & Denis, V. (2020). Stranger things: organismal traits of two octocorals associated with singular Symbiodiniaceae in a high-latitude coral community from northern Taiwan. *Frontiers in Marine Science*, *7*(1158). https://doi.org/10.3389/fmars.2020.606601

2 under review in scientific journals

Hsu, T.-H. T., Hoey, A. S., Clark, G. F., Ferrari, R., Fournet, C., Marrable, I., Piccaluga, A., Turnbull, J. W., Wilson, B., & Figueira, W. F. (in review). Understanding the drivers of coral consumption rates by butterflyfishes in a lagoonal reef system.

Marrable, I., Piccaluga, A., **Hsu, T.-H. T.**, Turnbull, J., Warren, C., Marzinelli, E. M., Ortiz, J.-C., & Figueira, W. F. (in review). Dietary specialisation modulates vulnerability of butterflyfishes to disturbance events.

1 master's thesis